

Log File Analyzer for Intrusion Detection

Introduction:

System logs contain valuable information about user activity, system behavior, and potential security threats. Analyzing these logs helps identify suspicious patterns such as unauthorized access attempts and abnormal traffic. This project focuses on building a Log File Analyzer that detects intrusion attempts using pattern-based analysis.

Objectives:

The main objectives of this project are:

- To parse and analyze system and web server logs
- To detect brute-force login attempts
- To identify suspicious high-frequency traffic
- To generate a report of potential threats
- To provide a simple CLI-based analysis tool

Technologies Used:

- Python
- Regular Expressions (re)
- collections (defaultdict)

System Design:

The system processes log files and extracts relevant information such as IP addresses and activity patterns. It uses pattern matching techniques to identify suspicious behavior.

Log Parsing

- Extracts IP addresses from Apache logs
- Identifies failed login attempts from SSH logs

Data Storage

- Uses dictionaries to count requests and failed attempts

Detection Logic

- Applies threshold-based rules to identify anomalies

Key Features:

Brute Force Detection

Detects repeated failed login attempts from the same IP address, indicating possible password guessing attacks.

High Traffic Detection

Identifies IPs generating excessive requests, which may indicate scanning or denial-of-service attempts.

Report Generation

Generates a text report summarizing suspicious activity.

Input Handling

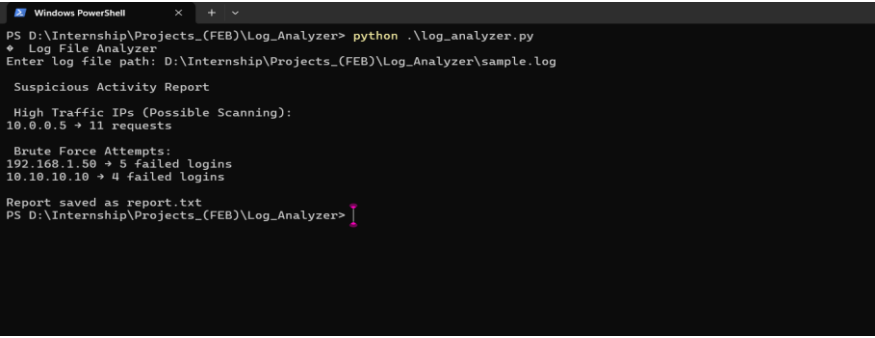
Supports flexible file path input for better usability.

Working Principle:

1. User provides log file
2. System reads log entries
3. Extracts IP addresses and activity
4. Counts requests and failed logins
5. Applies thresholds to detect anomalies
6. Displays and saves report

Results:

The tool successfully detects suspicious IPs involved in brute-force attacks and abnormal traffic patterns. It provides clear and readable output for analysis.



```
PS D:\Internship\Projects_(FEB)\Log_Analyzer> python .\log_analyzer.py
Log File Analyzer
Enter log file path: D:\Internship\Projects_(FEB)\Log_Analyzer\sample.log

Suspicious Activity Report

High Traffic IPs (Possible Scanning):
10.0.0.5 → 11 requests

Brute Force Attempts:
192.168.1.50 → 5 failed logins
10.10.10.10 → 4 failed logins

Report saved as report.txt
PS D:\Internship\Projects_(FEB)\Log_Analyzer>
```

Conclusion:

This project demonstrates how log analysis can be used for intrusion detection. It highlights the importance of monitoring system logs to identify potential security threats.